

Common Anchor

Bridge Equations to Observer Patch Holography and the Electroweak Scale as
Meeting Point

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Common Anchor: Bridge Equations to Observer Patch Holography and the Electroweak Scale as Meeting Point

Abstract. Observer Patch Holography (OPH) lists as an open item that length and tick do not follow from its own update rules. This document writes out the bridge to FFGFT as equations. The OPH cell energy is exactly the FFGFT bit energy $E_{\text{bit}} = \hbar c/L$ at the cell edge $L_{\text{cell}} = \sqrt{P} \ell_P$ **[B]**; the collapse threshold of Doc. 329 gives $n_{\text{thr}} = \sqrt{P/2}$ there, with the boundary at $P = 2$ **[B]**; $\tilde{T} \cdot m = 1$ supplies a native tick $\tilde{T}_{\text{cell}} = \sqrt{P} t_P$. Below ν , FFGFT delivers all masses as ratios to ν . The electroweak scale ν is therefore the meeting point: the conditional OPH hierarchy $\nu/E_\star$ **[Q]** and the FFGFT ladder m_e/ν **[K]** together give m_e/E_P to 0.93 %. A single SI anchor then fixes both frameworks.

List of symbols

- [B]** algebraically proven
- [K]** numerically confirmed
- [S]** structurally plausible, derivation open
- [X]** checked and negative
- [Q]** taken from an external source, cited

Symbol	Meaning
ξ	FFGFT scaling parameter (4/30000)
ℓ_P, t_P, E_P	Planck length, time, energy (not reduced)
$L_0 = \xi \ell_P$	FFGFT resolution floor (Doc. 180)
$E_{\text{bit}}(L) = \hbar c/L$	system-dependent bit energy (Doc. 257)
$n_{\text{thr}}(L)$	collapse threshold, number of bit quanta up to Compton = Schwarzschild (Doc. 329)
$M_{\text{coll}} = m_P/\sqrt{2}$	collapse mass (Doc. 329)
$\tilde{T}$	intrinsic time scale, $\tilde{T} \cdot m = 1$
ν	electroweak scale (246.22 GeV, from G_F)
r_i, p_i	prefactor and exponent of the ladder $m_i = r_i \xi^{p_i} \nu$
P	OPH pixel ratio a_{cell}/ℓ_P^2
$L_{\text{cell}}, E_{\text{cell}}$	OPH cell edge $\sqrt{P} \ell_P$, cell energy $E_P/\sqrt{P}$
$E_\star$	OPH reference energy of the hierarchy $\nu/E_\star$
α_U	OPH unified coupling
N	OPH capacity of the whole network

1 Occasion and classification

Tracking item #740 of the OPH repository records that length and tick do not follow from the OPH updates: length is tied to an outside value only through the capacity N and $\Lambda \ell_P^2 = 3\pi/N$, and the tick only through ratios. Doc. 364 examined the icosahedral OPH carrier spectrally; this document treats the question of scale.

The roles are not symmetric. FFGFT derives from ξ ; measured values and Standard Model quantities serve only for conversion and comparison. OPH quantities enter here as cited inputs from an external framework **[Q]** and carry the status their own documentation assigns them. The cosmic sector stays bracketed (P39); N is not used.

2 Inputs

FFGFT

- Bit energy $E_{\text{bit}}(L) = \hbar c/L$, system-dependent (Doc. 257).
- Collapse threshold $n_{\text{thr}}(L) = L/(\sqrt{2} \ell_P)$ and $M_{\text{coll}} = m_P/\sqrt{2}$ from Compton = Schwarzschild (Doc. 329) **[B]**.
- Resolution floor $L_0 = \xi \ell_P$ as a ratio (Docs. 180, 329).
- $\tilde{T} \cdot m = 1$: every energy has a proper time $\hbar/E$.
- Lepton ladder $m_e = \frac{4}{3} \xi^{3/2} \nu$ (Docs. 006, 373) **[K]**.

OPH [Q]

- Pixel ratio $P = a_{\text{cell}}/\ell_P^2$ as the fixed point of $P = \varphi + \sqrt{\pi}/A_T(P)$; public root $P = 1.630968\dots$, source root $1.630972\dots$; cell energy $E_{\text{cell}} = E_P/\sqrt{P}$ (code/P_derivation/FULL_DERIVATION.md). The public root contains an empirical hadronic contribution; the source root does not.
- Hierarchy $\nu/E_\star = P^{-1/2} \exp[-2\pi/(4\alpha_f(P))] = 2.01998 \times 10^{-17}$ (Theorem A, code/particles/hierarchy/theorem_package.md); explicitly conditional, the physical identity of $E_\star$ is open in OPH.

3 Bridge equations for length and tick

All statements in this section are dimensionless in Planck units; no SI anchor is consumed.

B1 — dictionary [B]. With $L_{\text{cell}} = \sqrt{P} \ell_P = 1.2771 \ell_P$,

$$E_{\text{cell}}^{\text{OPH}} = \frac{E_P}{\sqrt{P}} = \frac{\hbar c}{L_{\text{cell}}} = E_{\text{bit}}(L_{\text{cell}}). \quad (1)$$

Both frameworks use the same energy–length relation. OPH fixes the length by its fixed point; in FFGFT it stays system-dependent.

B2 — collapse threshold [B]. Inserted into Doc. 329:

$$n_{\text{thr}}(L_{\text{cell}}) = \sqrt{P/2} = 0.9030, \quad \frac{r_s(E_{\text{cell}})}{L_{\text{cell}}} = \frac{2}{P} = 1.2263. \quad (2)$$

The boundary above which a single cell quantum stays outside its own Schwarzschild radius is $P = 2$. The OPH root lies below it: a cell quantum localised on one cell sits at the collapse scale. For a holographic pixel this is expected rather than objectionable, but it is a sharp condition that any future change of P must respect or explain. Source and public roots differ here by 1.2×10^{-6} ; the statement does not depend on the open hadronic part.

B3 — tick. $\tilde{T} \cdot m = 1$ assigns the cell a proper time:

$$\tilde{T}_{\text{cell}} = \frac{\hbar}{E_{\text{cell}}} = \sqrt{P} t_P = \frac{L_{\text{cell}}}{c}. \quad (3)$$

As arithmetic this is **[B]**; physically it gives OPH a tick per cell in the same unit as its length, without a master clock. Tying it to a laboratory clock is possible but not yet carried out on the OPH side **[S]**. A laboratory clock, such as the caesium transition, is itself a matter system with energy ΔE_{clk} and, by $\tilde{T} \cdot m = 1$, its own tick $\hbar/\Delta E_{\text{clk}}$. The attachment requires the ratio $\Delta E_{\text{clk}}/E_{\text{cell}}$, computed within OPH itself; OPH lists this step as open (clock frontier $\varepsilon_{\text{clk}} = \Delta E_{\text{clk}}/E_*$, caesium route as an incomplete candidate). In FFGFT the question is not open: every mode has its tick $\tilde{T} = 1/m$, and tick ratios are mass ratios. FFGFT thus supplies the translation principle, not the missing OPH value.

B4 — floor [K]. $L_{\text{cell}}/L_0 = \sqrt{P}/\xi = 9578$. The OPH cell lies about four orders of magnitude above the FFGFT resolution floor; no conflict.

4 The meeting point ν

The two frameworks cover different sections of the scale ladder. OPH computes, conditionally, from the top down to ν ; FFGFT from ν downwards: all charged lepton masses via the ladder $m_i = r_i \xi^{p_i} \nu$ (Docs. 006, 373), the boson masses via $\sin^2 \theta_W = 2/9$ and the trace rule (Docs. 372, 373).

Quantity	Origin	vs. measured
$\nu/E_* = 2.01998 \times 10^{-17}$	OPH, Theorem A [Q]	+0.16 % (with $E_* = E_P$)
$m_e/\nu = \frac{4}{3} \xi^{3/2} = 2.0528 \times 10^{-6}$	FFGFT, Docs. 006/373 [K]	−1.09 % (bare)
$m_e/E_P = 4.1466 \times 10^{-23}$	product	−0.93 %

For the reference energy E_* see the following subsection. The −1.09 % on the FFGFT side is the known gap between bare and measured lepton masses (Doc. 352), not a new finding. The product depends on the OPH branch only at the 10^{-4} level.

Consequence: a single SI anchor fixes both frameworks. With G as anchor, the OPH hierarchy leads to ν , and the FFGFT ladder leads from there to all particle masses; with m_e as anchor, the chain runs the other way. The residual of about 1 % is the common consistency test. FFGFT also reaches m_e/E_P through its own chain; the meeting point ν is thus a comparison of two independent routes.

The reference energy E_*

What E_* is. In OPH, E_* is the energy unit of the calculation. The code sets the Planck energy equal to one; the cell energy is $E_P/\sqrt{P}$, the unification scale $M_U = E_P e^{-2\pi P^{1/6}}$, the hierarchy $\nu/E_* = P^{-1/2} \exp[-2\pi/(4\alpha_U)]$ (reproduced, check script D1). Which physical energy this unit is, OPH explicitly leaves open; it is to be fixed through a clock, $E_* = \hbar \nu_{\text{clk}}/\varepsilon_{\text{clk}}$, that is, through the same open step as in B3.

Two candidates. The Planck energy exists in two conventions: non-reduced $E_P = \sqrt{\hbar c^5/G} = 1.2209 \times 10^{19}$ GeV and reduced $\tilde{E}_P = E_P/\sqrt{8\pi} = 2.435 \times 10^{18}$ GeV. They differ by $\sqrt{8\pi} \approx 5.01$, depending on whether the gravitational coupling is written as G or as $8\pi G$. With measured ν , OPH gives $E_\star = 1.2189 \times 10^{19}$ GeV: 0.16 % below E_P , but a factor of 5.0 above $\tilde{E}_P$ **[K]**. The reduced reading is thereby excluded; the non-reduced one fits.

How sharp 0.16 % is. α_U sits in the exponent. The sensitivity is

$$\frac{\partial \ln(\nu/E_\star)}{\partial \alpha_U} = \frac{\pi}{2\alpha_U^2} \approx 929. \quad (4)$$

The deviation of 0.16 % corresponds to a shift of α_U by 1.7×10^{-6} , relatively 4×10^{-5} **[B]**. The same sensitivity accounts exactly for the difference between the public and the source branch (-8.4×10^{-5} , check script D5). The agreement with E_P is therefore not a rough match of orders of magnitude but a tight condition on α_U .

Why a reading [S]. OPH itself does not claim $E_\star = E_P$; the assignment here follows from the numerical comparison. It would be confirmed if OPH closes the clock attachment and obtains $E_\star = E_P$, or if OPH singles out the coupling without 8π structurally. For the comparison in §5 the reading is not critical: it only fixes against which of the two Planck energies the 0.16 % is measured.

5 Two routes to m_e/E_P and their accuracy

FFGFT reaches the Planck scale through its own chain (Docs. 012, 013, 180):

1. $G = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} = 6.6745 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$, +0.003 % against CODATA. The notation in Doc. 180 splits the same value into $C_{\text{dim}} \cdot C_{\text{conv}}$; both forms are equivalent.
2. $\ell_P = \sqrt{\hbar G/c^3} = 1.6163 \times 10^{-35} \text{ m}$ and hence $L_0 = \xi \ell_P = 2.155 \times 10^{-39} \text{ m}$.
3. $E_P = 1.22087 \times 10^{22} \text{ MeV}$, $m_e/E_P = 4.1855 \times 10^{-23}$ (+0.002 %) **[K]**.
4. $\nu/E_P = 2.01676 \times 10^{-17}$.

Not every route of calculation yields the same accuracy:

Route	Quantity	Dev.	Origin of the deviation
FFGFT chain $\xi \rightarrow G \rightarrow E_P$	m_e/E_P	+0.002 %	accuracy of G
FFGFT ladder	m_e/ν	−1.09 %	bare versus measured (Doc. 352)
OPH, Theorem A [Q]	$\nu/E_\star$ vs. FFGFT chain	+0.16 %	reading $E_\star = E_P$, conditional fixed point
Combination via ν	m_e/E_P	−0.93 %	sum of the two preceding

The routes do not contradict each other. Each deviation is assigned to a known place: the bare residual of the lepton ladder on one side, the identification of $E_\star$ on the other. For the comparison table, the route by which a value was obtained must therefore always be stated alongside the value.

6 What the bridge does not achieve

- The upper part of the chain comes from an external framework and is conditional there **[Q]**; the combination is a consistency test between the frameworks.

- Unit of time: in FFGFT every time scale follows from $\tilde{T} = 1/m$; the SI second serves only for conversion. In OPH the unit of time is tied to $E_\star$ and is to be fixed through the clock attachment; this step is open on the OPH side **[S]**.
- N and the cosmic sector are not involved (P39).
- The algebraic carriers still differ (A_5 versus $T^4/\mathbb{Z}_3$, Doc. 364); the bridge concerns scales, not symmetries.

Tool and method

Claude (Anthropic) was used as a tool for: reading the OPH source files, writing and running the Python check script, drafting the $\text{\LaTeX}$ sources, and maintaining register and changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every statement is secured by assertions in the check script. OPH values are taken verbatim from the public repository; measured values (CODATA 2022, PDG) serve only as comparators.

7 Result

1. B1: $E_{\text{cell}}^{\text{OPH}} = E_{\text{bit}}(\sqrt{P}\ell_P)$ **[B]**.
2. B2: $n_{\text{thr}}(L_{\text{cell}}) = \sqrt{P/2}$, boundary $P = 2$, OPH below it **[B]**.
3. B3: cell tick $\tilde{T}_{\text{cell}} = \sqrt{P}t_P$ from $\tilde{T} \cdot m = 1$; laboratory attachment open on the OPH side **[S]**.
4. B4: $L_{\text{cell}}/L_0 = \sqrt{P}/\xi \approx 9578$ **[K]**.
5. Meeting point ν : OPH $\nu/E_\star$ **[Q]** $\times$ FFGFT m_e/ν **[K]** gives m_e/E_P to -0.93% ; a single SI anchor suffices for both frameworks.
6. FFGFT's own chain $\xi \rightarrow G \rightarrow E_P$: m_e/E_P to $+0.002\%$ **[K]**; the accuracy depends on the route of calculation (§5).

Check script: 2/python/Dok374_Skripte/pruef_374_anker.py (27/27)

Register (Doc. 190): no entry.

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